(Please write your Enrollment Number) Enrollment No. 07401182021

END TERM EXAMINATION

(CBCS)(SUBJECTIVE TYPE)

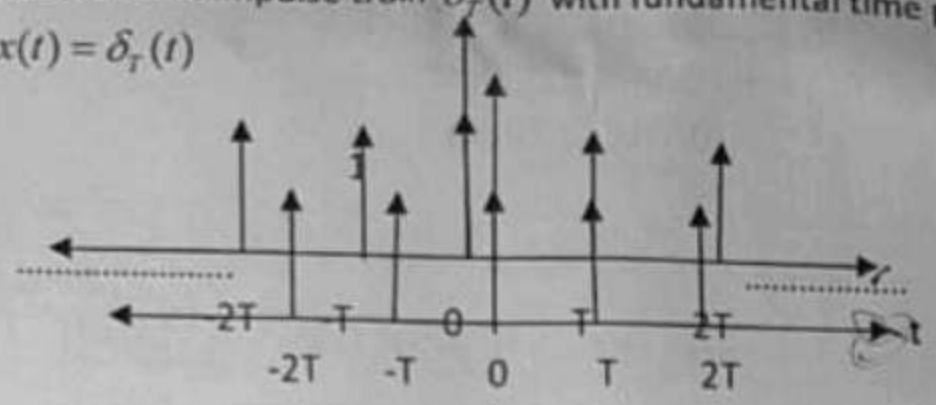
B. Tech (ECE-AI), 2nd Semester

(AUGUST, 2022)

Subject Code:BEC 106	Subject: Signal and Systems_CBCS
Time :3 Hours	Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1			(2.5*8=20)
	(a)	Explain the three properties of Convolution.	
	(b)	Distinguish between Deterministic and random signals along with examples.	
	(c)	What is odd symmetry? How does it help in simplification of calculations in Fourier series?	
	(d)	Distinguish between Fourier series and Fourier transform.	
	(e)	State and prove Modulation theorem for Discrete Time Fourier Transform.	
	(f)	Differentiate between phase delay and group delay.	
	(g)	When does aliasing occur? How it can be avoided.	
	(h)	What is ROC for Z-transform? Write any four properties of ROC.	
		UNIT-I	
Q2	a) Determine the response of the relaxed system characterized by the impulse response h(n) and input signal x(n) given below: $h(n)=\left(\frac{1}{2}\right)^n u(n)$ $x(n)=2^n u(n)$ b) Find the even and odd components of the following signal: $x(t) = 1+2t+3t^2+4t^3$		(5,5)
Q3	a) Determine whether the following system is time-invariant or not: $y(n)= x(n)+nx(n-2)$ Also, comment on its stability. b) What are basic operations on signals? Illustrate each with an example.		(5,5)
		UNIT-II	
Q4	a) Find the inverse Fourier transform of $X(w)=\frac{jw}{(2+jw)^2}$, given $F[te^{-at}u(t)]=\frac{1}{(a+jw)^2}$ and F is the Fourier transform. b) Find the Fourier transform of $x(t)=f(t-2)+f(t+2)$.		(5,5)

Q5	Find complex exponential Fourier series and the trigonometric Fourier series of unit impulse train $\delta_T(t)$ with fundamental time period T. $x(t) = \delta_T(t)$  	(10)
	UNIT-III	
Q6	a) Using the property of convolution for DTFT find the output y(n), when $h(n) = \left(\frac{1}{3}\right)^n u(n)$ and $x(n) = \left(\frac{1}{2}\right)^n u(n)$. b) Find the frequency response of the following causal system: $y(n) - x(n) - y(n-1) + \frac{3}{2}x(n-1) + \frac{6}{19}y(n-2) = 0$	(7,3)
Q7	a) The impulse response of a system is $h(n) = \begin{cases} 1 & \text{for } 0 \le n \le N\text{-}1 \\ 0 & \text{otherwise} \end{cases}$ Find the transfer function, frequency response, magnitude and phase response. b) Find DTFT of $x(n) = \left(\frac{1}{2}\right)^{n-2} u(n-2)$.	(6,4)
	UNIT-IV	
Q8	a) Determine Nyquist rate and Nyquist interval for $x(t) = 1 + \cos 5000\pi t + \sin 4000\pi t + \cos 7000\pi t$ b) State and prove sampling theorem.	(5,5)
Q9	a) Find the Z-transform and ROC of $x(n) = a^n u(-n)$. b) Find the inverse Z-transform of the following function $X(z) = \frac{z^2}{z-1} + \frac{z^2}{z-3}$ for ROCs (a) $\lvert z \rvert > 3$, (b) $\lvert z \rvert < 1$, (c) $1 < \lvert z \rvert < 3$	(5,5)